Software Requirements Specification (SRS)

Student Record System

Project: Student Record System

Version: 1.0

Authors:

Upadrasta Naga Sarvani, PES2UG24CS567

Vinaayak G Dasika, PES2UG24CS588

Yaseen mulla, PES2UG24CS607

Eshwar B H, PES2UG25CS808

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Revision History

Version	Date	Author	Change summary	Approval
1.0	06-09-2026	Team 7	Initial version of SRS	Pending

Approvals

Role	Name	Signature / Email	Date
Course Coordinator / Instructor			

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1. Introduction

1.1 Purpose

This SRS describes what our Student Record System is meant to do. The system helps the college keep track of student details, attendance, and marks, and gives students a simple way to check their own academic info online.

1.2 Scope

The system lets Admin staff manage student records, lets Faculty mark attendance and enter marks, and lets Students view their own profile, attendance and grades. It does not cover things like fee payment, hostel allocation, or library management — those are out of scope for this project.

1.3 Audience

Our project team (developers/testers), our course instructor/evaluator, and the people who would actually use the system — students, faculty and admin staff.

1.4 Definitions / Abbreviations

RBAC – Role Based Access Control. GPA – Grade Point Average. UI – User Interface. API – way the frontend and backend talk to each other. TLS/HTTPS – secure connection used when sending data over the internet. RTM – Requirements Traceability Matrix, a table that links requirements to test cases.

2. Overall description

2.1 Product perspective

This is a standalone web app — a frontend that runs in the browser, a backend server, and a database to store all the student data.

2.2 Main functions

- Login with role-based access (Student / Faculty / Admin)
- Add, edit and search student records
- Mark attendance and calculate attendance %
- Enter marks and calculate GPA
- Generate a basic mark sheet
- Simple admin-level reports
- Daily backup of the database

2.3 User roles

- Student — views their own profile, attendance and marks.
- Faculty — enters attendance and marks for the subjects they teach.
- Admin — manages student records and can view overall reports.

2.4 Operating environment

Backend runs on a normal server (e.g. Node.js/Express) with a MySQL or similar database. Frontend works in any modern browser (Chrome/Firefox/Edge), desktop or mobile.

2.5 Constraints

- Built for a single college's data, not multiple institutions.

3. External interface requirements

3.1 User interfaces

A simple web dashboard, different for each role — Student sees their own profile/attendance/marks, Faculty sees a page to enter attendance/marks, Admin sees student management and basic reports.

3.2 Hardware interfaces

No special hardware needed. Only a computer or phone with a browser and internet/college Wi-Fi.

3.3 Software interfaces

- Database: MySQL (or similar) accessed through the backend.

3.4 Communications

The frontend and the backend communicate over a REST API

4. System features (detailed)

4.1 Login & Access

Req ID	Requirement (shall…)	Type	Priority	Source	Acceptance criteria / Test ref	Notes
SRS-F-001	The system shall let a user log in as a Student, Faculty, or Admin, and shall only show that user the pages/data allowed for their role.	Functional	High	Team	Logging in as each role shows only that role's dashboard. TC1	Basic RBAC
SRS-F-002	The system shall mask password characters while the user is entering their password.	Functional	High	Team	DB check shows password field is hashed, not readable. TC2	
SRS-F-003	The system shall lock the account for a few minutes after 3 wrong password attempts in a row.	Functional	Medium	Team	4th wrong attempt shows a lockout message. TC3	-
SRS-F-004	The system shall log a user out automatically if they are inactive for a set amount of	Functional	Medium	Team	Session ends and redirects to login after idle time. TC4	-

Req ID	Requirement (shall...)	Type	Priority	Source	Acceptance criteria / Test ref	Notes
	time (e.g. 15 min).					

4.2 Student Records

Req ID	Requirement (shall...)	Type	Priority	Source	Acceptance criteria / Test ref	Notes
SRS-F-010	The Admin shall be able to add a new student record (roll no, name, branch, semester, contact info).	Functional	High	Admin	New record shows up when searched. TC5	Roll no must be unique
SRS-F-011	The Admin shall be able to edit an existing student's details.	Functional	High	Admin	Edited fields are saved and shown correctly on next view. TC6	-
SRS-F-012	The Admin shall be able to mark a student record as inactive instead of deleting it outright.	Functional	Medium	Admin	Inactive students don't show in the active list but data isn't lost. TC7	So we don't lose old data
SRS-F-013	A logged-in Student shall be able to view their own profile, attendance and marks.	Functional	High	Student	Student sees only their own info, not other students'. TC8	-
SRS-F-014	The Admin/Faculty shall be able to search students by roll number, name, branch or semester.	Functional	Medium	Admin/ Faculty	Search returns the correct matching record(s). TC9	-

4.3 Attendance & Marks

Req ID	Requirement (shall...)	Type	Priority	Source	Acceptance criteria / Test ref	Notes
SRS-F-020	Faculty shall be able to enter/update marks for a subject they teach.	Functional	High	Faculty	Entered marks are saved and visible to that student only. TC10	-
SRS-F-021	The system shall calculate a student's GPA automatically from their subject marks.	Functional	High	Team	Calculated GPA matches a manually worked-out example. TC11	
SRS-F-022	The system shall generate a basic mark sheet for a student that can be downloaded/printed.	Functional	Medium	Student	Downloaded mark sheet shows correct subject-wise marks. TC12	PDF or printable HTML page
SRS-F-023	Faculty shall be able to mark attendance for a class, and the system shall calculate	Functional	High	Faculty	Attendance % updates correctly after new entries. TC13	-

Req ID	Requirement (shall…)	Type	Priority	Source	Acceptance criteria / Test ref	Notes
	attendance percentage.					

4.4 Reports & Backup

Req ID	Requirement (shall…)	Type	Priority	Source	Acceptance criteria / Test ref	Notes
SRS-F-030	Admin shall be able to view a simple report of attendance/marks for a branch or class.	Functional	Low	Admin	Report totals match the individual student records. TC14	Basic version — no charts needed
SRS-F-040	The system shall back up the database at least once a day.	Functional	Medium	Team	Backup file is created daily and can be restored. TC15	

5. Non-functional requirements (detailed)

Req ID	Requirement	Category	Priority	How we'll check it
SRS-NF-001	The system shall load a student profile or return search results within 2 seconds for a normal class-size dataset.	Performance	Medium	Manual timing during testing
SRS-NF-002	The system should be usable throughout a normal class demo/session without crashing.	Reliability	Medium	Testing during dev + demo run
SRS-NF-003	Student personal and academic data should not be visible to users outside their role, and passwords must never be stored in plain text.	Security	High	Manual check of DB and role tests
SRS-NF-004	Marks and attendance data, once entered, should not be lost between sessions (persisted in the database, not memory).	Reliability/ Data	High	Restart server and confirm data is intact
SRS-NF-005	The interface should be usable on a laptop and on a phone browser without major layout issues.	Usability	Low	Manual check on 2-3 screen sizes

5.1 Security

5.1.1 Security objectives

- Keep student data private — only the right role should be able to see it.
- Make sure only logged-in, authorized users can change records (so marks/attendance can't be tampered with).

5.1.2 Security requirements

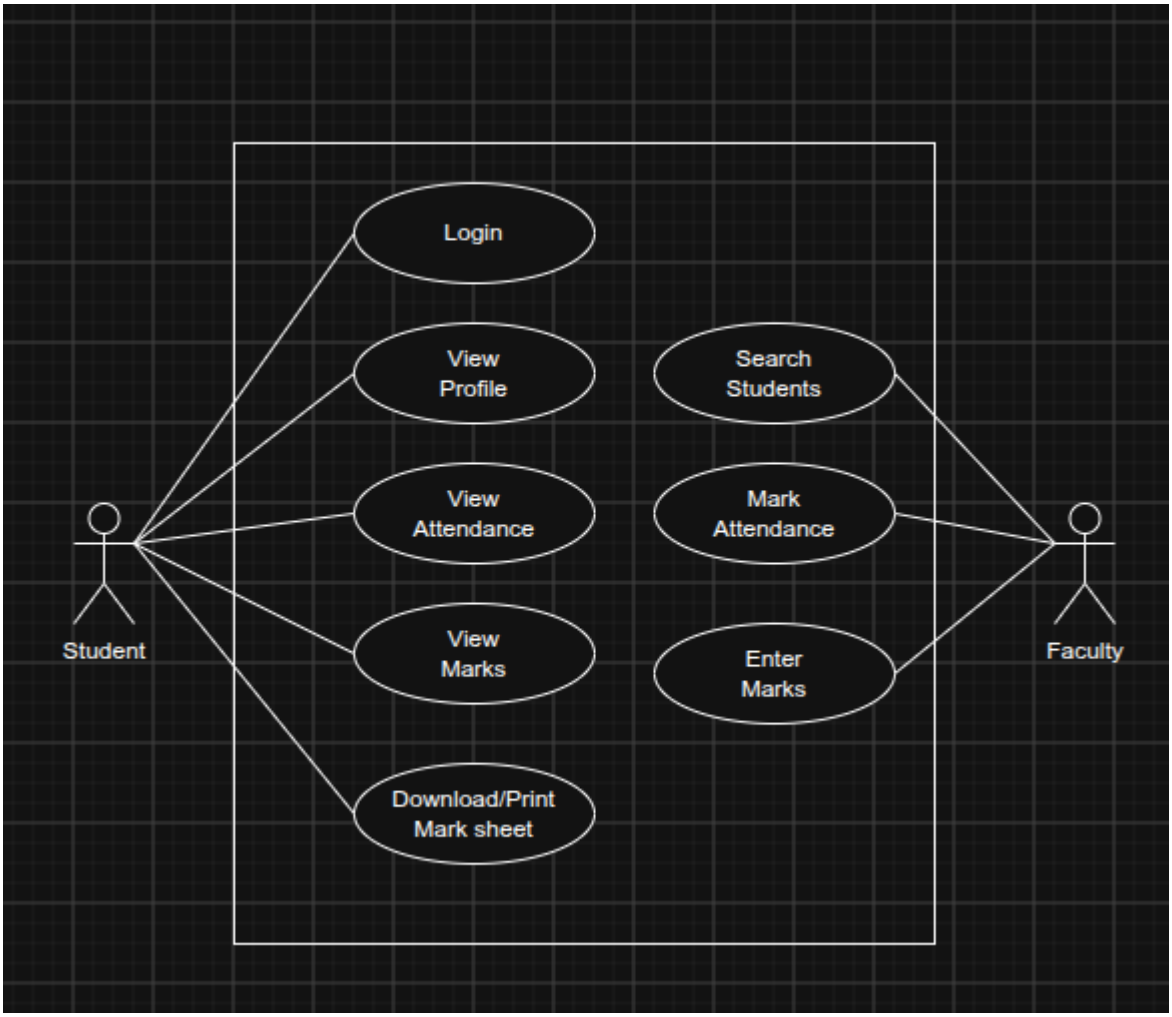
Req ID	Requirement (shall…)	Type	Priority	Source	Acceptance criteria / Test ref	Notes
SRS-SR-001	The system shall use HTTPS for communication between the browser and server.	Security	Medium	Team	Checked in browser dev tools during testing.	-
SRS-SR-002	Passwords must be stored as hashed values, never as plain text.	Security	High	Team	DB inspection shows only hashes stored.	-
SRS-SR-003	A user should not be able to access another role's pages just by guessing the URL — the server should also check the role, not just hide the button in the UI.	Security	High	Team	Try accessing an admin page as a student user; should be blocked.	-
SRS-SR-004	Any change to a student's marks or record should be traceable to which user made it, in case something needs to be corrected later.	Security	Medium	Team	Basic log/table showing last-edited-by for a record.	-
SRS-SR-005	A user's login session should expire after some time of inactivity, so an unattended logged-in browser can't be misused.	Security	Medium	Team	Leave a session idle past the timeout and confirm re-login is required.	-

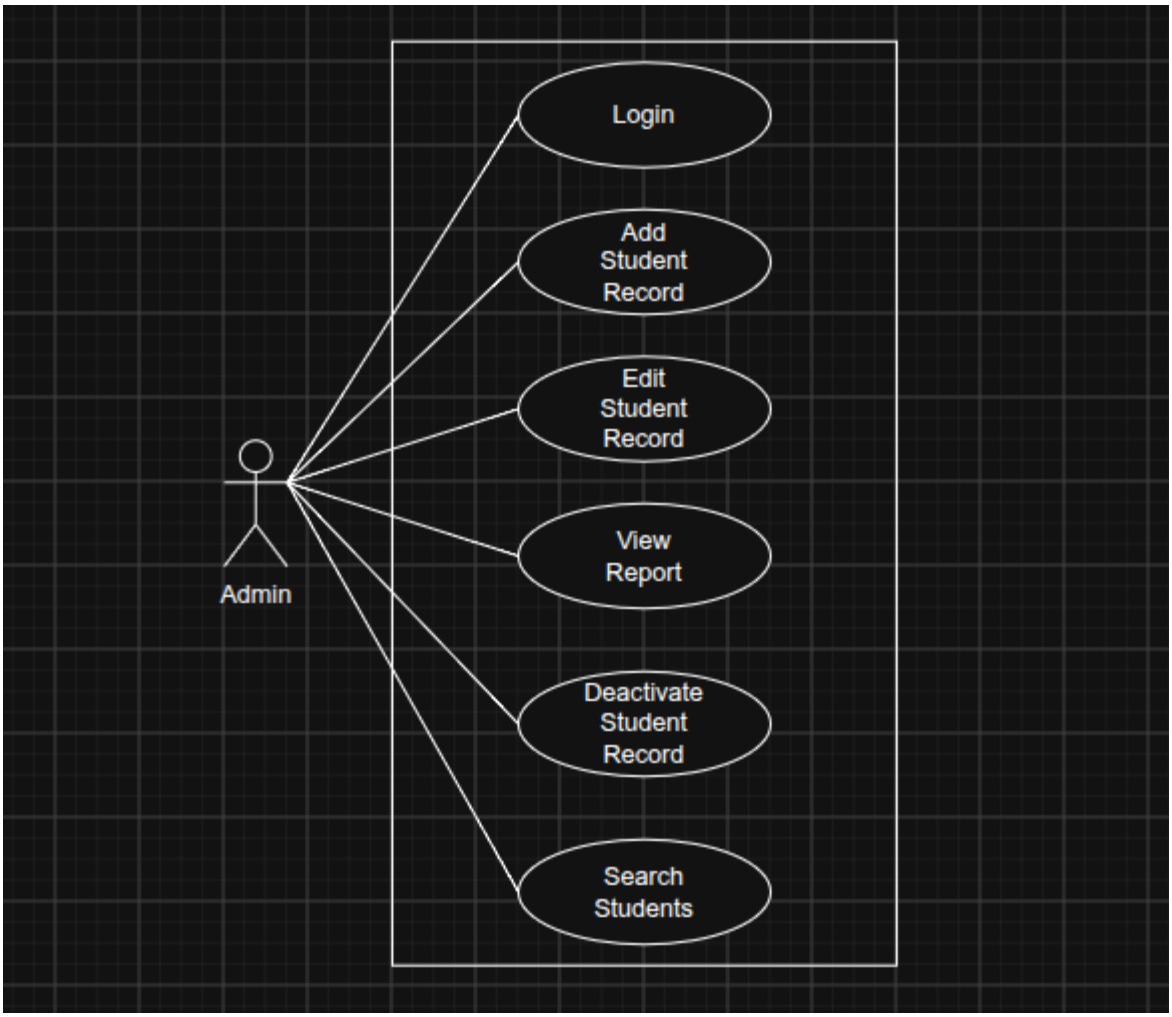
6. Quality attributes & Acceptance tests

- The project will be considered as completed when all High-priority requirements above are working and the RTM below is filled in with pass/fail for each.
- Things to test: login/roles, adding & searching records, attendance marking, marks entry + GPA calculation, and basic security checks (like role access).

7. System models and diagrams

7.1 UML Use-Case diagram





8.Requirements Traceability Matrix (RTM)

Req ID	Requirement (short)	Module	Test case	Status
SRS-F-001	Role-based login	Auth	TC1	Not started
SRS-F-002	Password masking	Auth	TC2	Not started
SRS-F-003	Lock account after failed logins	Auth	TC3	Not started
SRS-F-004	Auto logout after inactivity	Auth	TC4	Not started
SRS-F-010	Add student record	Records	TC5	Not started
SRS-F-011	Edit student record	Records	TC6	Not started
SRS-F-012	Deactivate student record	Records	TC7	Not started
SRS-F-013	View own profile (student)	Records	TC8	Not started
SRS-F-014	Search students	Records	TC9	Not started
SRS-F-020	Enter marks	Academics	TC10	Not started
SRS-F-021	Calculate GPA	Academics	TC11	Not started
SRS-F-022	Generate mark sheet	Academics	TC12	Not started
SRS-F-023	Mark attendance	Academics	TC13	Not started
SRS-F-030	View branch/class	Reports	TC14	Not started

Req ID	Requirement (short)	Module	Test case	Status
	report			
SRS-F-040	Daily database backup	Records	TC15	Not started
SRS-NF-001	Fast search/profile load	Records	-	Not started
SRS-NF-002	System stays usable in a session	System	-	Not started
SRS-NF-003	Data privacy by role	Auth/Records	-	Not started
SRS-NF-004	Marks/attendance data persisted	Academics	-	Not started
SRS-NF-005	Usable on laptop & phone	UI	-	Not started
SRS-SR-001	HTTPS for data in transit	Security/Auth	-	Not started
SRS-SR-002	Password hashing	Auth	-	Not started
SRS-SR-003	Server-side role check	Security/Auth	-	Not started
SRS-SR-004	Track who edited a record	Security/Auth	-	Not started
SRS-SR-005	Session expires after inactivity	Auth	-	Not started